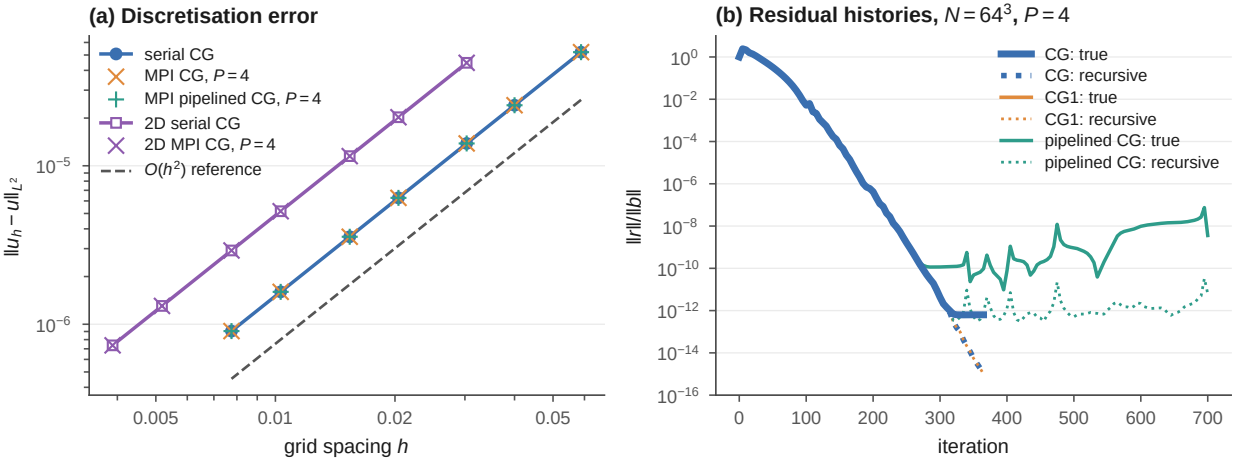
(a) Discretisation error
serial CG
MPI CG, P = 4
MPI pipelined CG, P = 4
2D serial CG
2D MPI CG, P = 4
O(h²) reference
||u_h − u||_{L²}
10⁻⁵
10⁻⁶
0.005
0.01
0.02
0.05
grid spacing h
(b) Residual histories, N = 64³, P = 4
CG: true
CG: recursive
CG1: true
CG1: recursive
pipelined CG: true
pipelined CG: recursive
||r||/||b||
10⁰
10⁻²
10⁻⁴
10⁻⁶
10⁻⁸
10⁻¹⁰
10⁻¹²
10⁻¹⁴
10⁻¹⁶
0
100
200
300
400
500
600
700
iteration